

InLine

Full Product Requirements, Page Inventory, Button Map & Codex Build Prompt

Tagline: Stand in line without standing there.

Field	Details
Product Type	SaaS queueing platform + virtual ticketing + SMS reminders + embeddable widgets
Primary Stack	React + TypeScript + Node.js + PostgreSQL + Prisma + Socket.IO
Target Market	Companies, clinics, banks, salons, government offices, schools, service centers, restaurants, and any place without a ticket system
Main Business Model	Monthly subscriptions + SMS wallet + add-ons + enterprise invoices + future runner marketplace commission
Prepared For	Shikulo Thomas / Edge work concept
Version	v1.0 - Product requirements and Codex prompt

One-line vision

InLine should make any business look like it has a modern ticketing system in one day - without expensive hardware, without app downloads, and without forcing people to physically wait in crowded lines.

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1. Executive Summary

InLine is a full-stack SaaS queueing system that helps businesses replace manual waiting lines, paper lists, WhatsApp queue messages, and expensive ticket machines with a simple web-first solution. Companies can create a public InLine company page, manage branches and services, generate QR codes, embed queue widgets into their own websites, and send SMS reminders to customers.

The platform targets companies that do not already have a ticket system. The product should be simple enough for a small salon to start using in minutes, but powerful enough for clinics, banks, schools, government offices, and multi-branch service companies.

The most important product promise is: customers should be able to stand in line without physically standing there. They join a queue from a phone, get a ticket number, track their position, receive SMS alerts, and arrive when their turn is close.

InLine also includes a future queue assistant marketplace: verified people can physically stand in line on behalf of users at places where InLine is not yet installed. This must be ethical, transparent, and rule-following: no queue cutting, no fake reservations, and no breaking the rules of the location.

Build philosophy

Make the MVP usable immediately, but design the architecture like a serious SaaS: multi-tenant companies, branches, services, roles, subscriptions, SMS wallet, real-time queue updates, API-ready structure, and embeddable widgets.

Core Outcomes

- Customers spend less time physically waiting.
- Businesses reduce crowding and improve service flow.
- Staff get a simple Call Next dashboard instead of paper lists.
- Managers get analytics on wait times, peak hours, staff load, and service demand.
- Companies can adopt InLine without installing expensive hardware.
- InLine can monetize through subscriptions, SMS credits, add-ons, enterprise billing, and future runner marketplace fees.

2. Product Vision and Differentiation

Most queueing tools stop at ticket numbers. InLine should go further: it should become a queue operating system for companies and a waiting-time assistant for customers. The product must feel premium, fast, and easy. It must also be flexible enough to live on InLine pages, company websites, QR posters, TV screens, and mobile browsers.

Differentiation

Differentiator	Requirement
No app required	Customers can join via mobile browser, SMS link, QR code, or embedded widget.
Embed-first	Companies can put their queue directly on their own website using iframe or JavaScript embed.
SMS-first for Namibia/local markets	Customers do not need push notifications or accounts; phone number and SMS can be enough.
Runner marketplace future	Users can request a verified assistant to stand in line at unsupported locations.
Local business friendly	Manual invoice/EFT payments and local payment adapters

Differentiator	Requirement
	make adoption easier.
Hardware optional	TV mode and kiosk mode are available, but the system works without them.
Real-time	Staff and customers see live updates through Socket.IO.
Multi-tenant SaaS	Each company has isolated data, roles, branches, services, billing, and branding.

Product Modules

Module	Purpose
Public Website	Marketing, pricing, product walkthrough, company directory, lead capture.
Customer Portal	Join queue, track ticket, get SMS, manage ticket.
Company Admin	Setup company, branches, services, staff, billing, embeds, analytics.
Staff Dashboard	Operate queues, call next, transfer, serve, miss, hold.
Super Admin	Manage all companies, plans, billing, platform settings, support.
Embed Widget	Company website queue widget and public join flow.
SMS Engine	Reminder templates, provider adapters, wallet billing, logs.
Billing Engine	Plans, subscriptions, invoices, payments, add-ons, trials.
Runner Marketplace	Future module for physical queue assistants at unsupported places.

3. Target Users and Permissions

Every action must be permission-controlled. Do not hardcode role checks randomly inside UI components only. Enforce role permissions on the backend and hide unavailable buttons on the frontend.

Role	Who	Main Permissions
Super Admin	InLine platform owner	Manage all companies, subscriptions, plans, invoices, SMS providers, global settings, support tickets, feature flags.
Company Owner	Business owner	Full control over own company, billing, staff, branches, services, settings, embeds, analytics.
Company Manager	Branch or company manager	Manage branches, staff, queues, services, reports. Limited billing access depending on permission.
Staff / Operator	Front desk person	View assigned branch queues, call next, mark served/missed, transfer tickets, send manual messages.
Customer	Person joining a queue	Join queue, view own ticket, cancel, mark on the way, rate service, save favorites if account exists.
Runner / Queue Assistant	Verified person who stands in line for users	Accept runner jobs, update queue position, check in, upload proof, complete job.
Support Agent	InLine customer support	View support-safe company details, impersonation with audit log, help resolve issues.

Permission Rules

- A staff user can only see queues for assigned company/branch unless given wider access.
- A company manager cannot access another company tenant.
- Public routes can read company profile and queue configuration, but cannot read private customer lists.
- Only authorized staff can call, serve, miss, cancel, hold, or transfer tickets.
- Only company owner or billing manager can change plan, buy SMS credits, or download invoices.
- Only super admin can create global plans, mark manual payments paid, or override billing status.
- All billing, queue state, staff, and SMS actions must create AuditLog entries.

4. Full Page Inventory

This is the required page map. Codex should create routes, responsive layouts, empty states, loading states, error states, and permission-based redirects for all pages below.

Public Marketing Website

Route	Page	Purpose
/	Home / landing page	Explain product, show live product cards, CTA to Start Free and Book Product Walkthrough.
/how-it-works	How It Works	Explain customer, staff, company, and runner flows.
/for-companies	For Companies	Explain business benefits, use cases, dashboard screenshots.
/for-customers	For Customers	Explain join queue, SMS reminders, ticket tracking.
/for-runners	For Queue Assistants	Future page explaining verified runners and safety rules.

Route	Page	Purpose
/pricing	Pricing	Free, Starter, Business, Enterprise, SMS credits, add-ons.
/product-tour	Product Tour	Production-like company queue flow with seeded validation data.
/directory	Company Directory	Search companies using InLine.
/contact	Contact / Book Product Walkthrough	Lead capture form.
/privacy	Privacy Policy	User data and SMS consent.
/terms	Terms of Service	Use rules, billing rules, runner rules.
/status	System Status	Future status page for uptime incidents.

Authentication and Onboarding

Route	Page	Purpose
/login	Login	Email/phone login for users, staff, managers, super admin.
/register	Register	Customer or company signup starting point.
/register/company	Company Signup	Create company owner account and company profile.
/register/customer	Customer Signup	Optional customer account registration.
/forgot-password	Forgot Password	Request reset link or OTP.
/reset-password	Reset Password	Set new password.
/verify-phone	Phone Verification	OTP phone verification.
/onboarding/company	Company Onboarding Wizard	Set company details, branch, services, plan, QR code.

Public Company Pages

Route	Page	Purpose
/c/:companySlug	Company Public Page	Company profile, branches, services, wait times, join queue.
/c/:companySlug/:branchSlug	Branch Public Page	Branch info, services, hours, queue stats.
/c/:companySlug/:branchSlug/:serviceSlug	Service Public Page	Specific service queue, join button, appointment option.
/ticket/:ticketId	Ticket Tracking Page	Live ticket status, people ahead, ETA, actions.
/join/:companySlug	Fast Join Page	Mobile-first direct join from QR code.
/rating/:ticketId	Service Rating Page	Post-service rating and feedback.

Customer Dashboard

Route	Page	Purpose
/app	Customer Home	Active tickets, favorites, recent companies.
/app/tickets	My Tickets	Current and past tickets.
/app/favorites	Favorite Companies	Saved companies and branches.
/app/profile	Customer Profile	Name, phone, email, notification preferences.
/app/runner-requests	Runner Requests	Future list of queue assistant requests.

Company Admin Dashboard

Route	Page	Purpose
/dashboard	Company Overview	Stats, live queue summary, setup checklist.
/dashboard/setup	Setup Checklist	Guided setup tasks.
/dashboard/branches	Branches	Create and manage branches.
/dashboard/branches/:id	Branch Detail	Branch services, counters, staff, queues.
/dashboard/services	Services	Create service types and average duration.

Route	Page	Purpose
/dashboard/counters	Counters / Windows	Manage service counters.
/dashboard/staff	Staff	Invite and manage staff roles.
/dashboard/queues	Queues	All live queues and historical queue sessions.
/dashboard/appointments	Appointments	Future appointment scheduling.
/dashboard/customers	Customers	Customer history and ratings.
/dashboard/sms	SMS Center	Templates, logs, balance, top-ups.
/dashboard/analytics	Analytics	Wait times, service time, peak hours, no-shows.
/dashboard/reports	Reports	Export PDF/CSV reports.
/dashboard/embed	Embed Widgets	Iframe and JS widget settings.
/dashboard/qr-codes	QR Codes	Generate and download QR posters.
/dashboard/branding	Branding	Logo, colors, custom page settings.
/dashboard/billing	Billing	Plan, subscription, invoices, payments, SMS credits.
/dashboard/settings	Company Settings	General settings, queue rules, security.
/dashboard/api-keys	API Keys	Future API access for integrations.

Staff / Operator App

Route	Page	Purpose
/staff	Staff Home	Assigned branches and counters.
/staff/queue	Live Queue Console	Call Next, current ticket, waiting list.
/staff/ticket/:id	Ticket Detail	Ticket actions and customer notes.
/staff/tv	TV Mode	Full-screen display for waiting room.
/staff/kiosk	Kiosk Mode	Walk-in self-service join screen.

Embed and Widgets

Route	Page	Purpose
/embed/:companySlug	Embed Widget Page	Iframe-safe widget.
/embed/:companySlug/:branchSlug	Branch Embed	Branch-specific widget.
/embed/ticket/:ticketId	Ticket Embed	Tiny status card.
/widget.js	JavaScript Widget Script	Injectable widget loader.

Super Admin

Route	Page	Purpose
/admin	Platform Overview	Companies, revenue, SMS, health.
/admin/companies	Companies	Manage tenants.
/admin/companies/:id	Company Detail	Support and billing controls.
/admin/plans	Plans	Manage pricing plans and limits.
/admin/invoices	Invoices	Manual invoice management.
/admin/payments	Payments	Payment logs and provider webhooks.
/admin/sms-providers	SMS Providers	Provider adapters and logs.
/admin/support	Support Inbox	Support requests.
/admin/feature-flags	Feature Flags	Enable beta features per company.
/admin/audit-logs	Audit Logs	Search platform actions.
/admin/settings	Platform Settings	Global configuration.

5. Button Inventory by Page

Every page must have clear primary and secondary actions. Buttons must show loading state, disabled state, error feedback, and success feedback where relevant.

Page/Area	Required Buttons	Notes
Home	Start Free, Book Product Walkthrough, View Pricing, Open Live Queue, For Companies, Watch Product Walkthrough, Contact Sales	CTA buttons should be visible above the fold.
Pricing	Start Free, Choose Starter, Choose Business, Contact Sales, Buy SMS Credits, Compare Plans	Plan selection must respect auth state.
Login	Login, Continue with Google optional, Forgot Password, Create Account, Show Password	Show invalid credentials error.
Company Signup	Create Company, Back, Continue, Save and Exit, Verify Phone	Wizard-style flow.
Company Public Page	Join Queue, Book Appointment, View Branches, Get Directions, Share, Call, View Wait Time	Must work for logged-out users.
Branch Page	Join Queue, Select Service, Get Directions, View Hours, Back to Company	Show closed state when branch is closed.
Service Page	Join This Queue, Book Appointment, View Ticket Rules, Back	Show average wait time.
Join Queue Form	Join Queue, Send OTP optional, Confirm, Cancel, Back, Clear Form	Phone number required.
Ticket Page	I am On My Way, Cancel Ticket, Rejoin if Allowed, Share Ticket, Get Directions, Refresh, Rate Service	Ticket page must update live.
Customer Home	Find Company, Scan QR, View Active Ticket, Save Favorite, Request Runner future	Mobile-first.
Company Overview	Create Branch, Create Service, Invite Staff, Generate QR, Open Staff Console, View Public Page	Also show setup checklist.
Branches	Add Branch, Edit, Disable, View Public Page, Generate QR, Assign Staff, Duplicate Branch	Never hard-delete active branch with tickets.
Services	Add Service, Edit, Disable, Reorder, Assign Branches, Set Average Time	Service can be global or branch-specific.
Counters	Add Counter, Edit, Disable, Assign Staff, Set Active/Inactive	Counter names appear in SMS.
Staff	Invite Staff, Resend Invite, Edit Role, Disable User, Reset Password, Assign Branch, Remove	Role-based access.
Queue Console	Call Next, Recall, Start Serving, Mark Served, Mark Missed, Put on Hold, Transfer, Cancel, Send SMS, Add Walk-in	Most important staff page.
Ticket Detail	Call, Serve, Miss, Hold, Transfer, Cancel, Add Note, Send SMS, View Events	Create audit events.
TV Mode	Fullscreen, Exit Fullscreen, Switch Branch, Switch Service, Theme, Refresh	Display-only but staff can configure.
Kiosk Mode	Start, Select Service, Enter Phone, Print/Show Ticket, Back, Help	For tablet at reception.
SMS Center	Buy Credits, Create Template, Edit Template, Send Test SMS, View Logs, Export Logs	Show low balance warnings.
Analytics	Date Range, Export CSV, Export PDF, Filter Branch, Filter Service, Compare Branches	Charts and summaries.
Embed Widgets	Copy iframe Code, Copy JS Code, Preview Widget, Add Allowed Domain, Save Settings, Regenerate Key	Embeds must be safe.
QR Codes	Generate QR, Download PNG, Download PDF Poster, Print, Copy Link, Regenerate	Separate QR per company/branch/service.
Branding	Upload Logo, Remove Logo, Pick Colors, Preview Page, Save Branding, Reset Theme	Use company colors on public pages and embed.
Billing	Upgrade, Downgrade, Cancel Plan, Renew, Download Invoice, Pay Invoice, Upload Proof, Buy SMS Credits	Manual and online payments.
Super Admin Companies	View, Suspend, Activate, Impersonate with Audit, Edit Billing, Create Invoice, Add Note	Use strong audit logging.
Super Admin Plans	Create Plan, Edit Plan, Archive Plan, Set Limits, Preview Pricing	Plans define feature gates.

Page/Area	Required Buttons	Notes
Runner Request Future	Request Runner, Estimate Price, Confirm Rules, Cancel Request, Chat, View Proof, Release Payment, Rate Runner	Future marketplace flow.

Global Button Rules

- Every destructive button must show a confirmation modal: Disable, Delete, Cancel Subscription, Cancel Ticket, Suspend Company.
- Every async button must show loading state and prevent double-click duplicates.
- Every form must have Save, Cancel/Back, and validation feedback.
- Primary CTA should be visually clear; secondary actions should not compete.
- Mobile staff console must keep Call Next and Current Ticket actions reachable without scrolling too much.
- Buttons hidden by permissions must also be blocked on backend; never rely on frontend only.

6. End-to-End User Flows

Customer joins queue from QR

1. Customer scans QR code at branch.
2. System opens /join/:companySlug with branch/service preselected.
3. Customer enters name and phone.
4. System creates QueueTicket and sends ticket-created SMS.
5. Ticket page opens and updates live.
6. When people ahead $\leq$ configured threshold, system sends almost-your-turn SMS.
7. Staff calls ticket; customer receives called SMS.
8. Staff marks served; customer gets rating page.

Company onboarding

9. Owner registers.
10. Creates company profile.
11. Adds first branch.
12. Adds services.
13. Invites staff.
14. Chooses Free/Trial plan.
15. Generates QR code.
16. Copies embed code or opens public page.
17. Starts taking queue entries.

Staff serves customer

18. Staff logs in.
19. Selects branch/counter.
20. Views waiting list.
21. Clicks Call Next.
22. System updates ticket to CALLED and emits real-time event.
23. Staff clicks Start Serving when customer arrives.
24. Staff clicks Served when done.
25. Queue analytics update.

Manual walk-in added by staff

26. Customer arrives physically.
27. Staff clicks Add Walk-in.
28. Staff enters name/phone/service.
29. System assigns ticket number.
30. Optional SMS is sent.
31. Customer waits or tracks link.

Company pays subscription manually

32. Company chooses plan.
33. System creates invoice.
34. Company pays by EFT or manual method.

- 35. Company uploads proof or admin verifies payment.
- 36. Super admin marks invoice paid.
- 37. Subscription becomes ACTIVE or MANUAL_ACTIVE.

Company buys SMS credits

- 38. Company opens SMS Center or Billing.
- 39. Clicks Buy Credits.
- 40. Selects bundle.
- 41. Pays online or by invoice.
- 42. SmsWallet balance increases after payment.
- 43. SMS sending deducts per message.

Runner future flow

- 44. User enters unsupported location and reason.
- 45. System estimates price and asks user to confirm rules.
- 46. Verified runner accepts job.
- 47. Runner checks in at location using GPS.
- 48. Runner updates queue position.
- 49. User receives SMS updates.
- 50. Runner completes job after handover.
- 51. Payment is released and both parties rate each other.

7. Queue Engine Requirements

The queue engine is the core of InLine. It must maintain fair ticket numbering, live positions, accurate enough estimated waiting times, and staff actions. It should work even when many customers join from QR, public page, embed, and staff console at the same time.

Requirement	Details
Ticket Numbering	Ticket numbers should be easy to read, e.g. A001, A002 per service or branch. Prevent duplicates by transaction/locking.
Daily Queues	Each branch/service should have queue sessions by date. Optionally reset numbers daily.
Ticket Sources	Track source: QR, public page, embed, staff walk-in, appointment, runner.
Position Calculation	People ahead should count WAITING/ON_HOLD rules correctly and exclude served/cancelled/missed.
ETA Calculation	Estimated wait = peopleAhead * rolling average service time, adjusted by active counters and staff speed.
Priority Queues	Support elderly, disabled, urgent, VIP, appointment priority. Must be transparent and configurable.
Fairness Rules	Prevent abuse by rate limiting repeated joins, optional OTP, and queue rule settings.
Closed Branch Logic	If branch is closed, joining should be disabled or scheduled for next opening depending on settings.
No-show Handling	Staff can mark missed. Company can allow rejoin within grace window.
Transfers	Move ticket from one service to another while preserving event history.
Hold	Put ticket on hold without losing it; staff must specify reason optionally.
Audit Trail	Every status change creates TicketEvent with actor and

Requirement	Details
	timestamp.

Ticket Status Logic

Status	Meaning	Customer/Staff Effect
WAITING	Ticket is in queue and waiting to be called.	Customer can view position and cancel.
CALLED	Ticket has been called to counter.	Customer sees counter and urgent status.
SERVING	Customer is being served.	Staff can mark served or transfer.
SERVED	Service completed.	Ticket leaves live queue; rating can be requested.
MISSED	Customer did not show.	Can optionally rejoin depending on company rules.
CANCELLED	Customer or staff cancelled.	Removed from queue.
TRANSFERRED	Moved to another service/queue.	Old ticket event history remains.
ON_HOLD	Temporarily paused.	Not counted same as waiting unless company setting says so.

8. SMS and Notification Requirements

SMS is a core requirement because customers should not need to install an app. The system must implement an adapter pattern so providers can be swapped without rewriting business logic.

SMS Provider Architecture

- Create SmsProvider interface with sendSms(to, message, metadata) and getDeliveryStatus(providerMessageId) if supported.
- Implement MockSmsProvider first for local development.
- Add placeholders for TwilioSmsProvider, AfricasTalkingSmsProvider, and LocalSmsProvider.
- Queue outgoing SMS jobs through BullMQ or an internal job queue, not inside request response where possible.
- Deduct SMS wallet balance only after successful send or reserve balance before send with rollback on failure.
- Every SMS must create Notification and SmsTransaction records.
- Support per-company templates and global fallback templates.
- Support opt-out rules and consent text in terms/queue forms.

Template Key	Message
TICKET_CREATED	Hi {{name}}, you joined {{company}} for {{service}}. Ticket {{ticketNumber}}. Track: {{ticketUrl}}
ALMOST_TURN	Hi {{name}}, your turn at {{company}} is coming soon. {{peopleAhead}} people ahead. Please start heading there.
CALLED	Hi {{name}}, ticket {{ticketNumber}} has been called at {{company}}. Go to {{counterName}}.
MISSED	Hi {{name}}, ticket {{ticketNumber}} was missed. Open your ticket to rejoin if allowed: {{ticketUrl}}
APPOINTMENT_REMINDER	Reminder: Your appointment at {{company}} is on {{date}} at {{time}}.
RUNNER_UPDATE	Your InLine runner update: {{message}}. Current position: {{position}}.
LOW_SMS_BALANCE	Your InLine SMS balance is low: {{balance}} credits. Top up to keep reminders active.

Notification Channels

Channel	Use
SMS	Primary notification channel for customers.
Email	Invoices, admin invites, reports, account recovery.
In-app/web notification	Live ticket status and dashboard alerts.
WhatsApp future	Optional provider adapter for markets where WhatsApp Business is useful.
Voice call future	Automated call for elderly customers or critical services.

9. Company Pages, Embeds and QR Codes

Embeds are a major advantage. Many companies already have websites and may want the queue on their existing site instead of sending customers to InLine only. Embeds must be easy to copy and safe to use.

Asset	Example	Requirement
Public page	https://inline.app/c/company-slug	Full page hosted by InLine.
Branch page	https://inline.app/c/company-slug/branch-slug	Branch-specific page.
Service page	https://inline.app/c/company-slug/branch-slug/service-slug	Service-specific join page.
iframe embed	<iframe src="https://inline.app/embed/company-slug?branch=branch-id" width="100%" height="600"></iframe>	Fastest MVP embed.

Asset	Example	Requirement
JS embed	<code><div id="inline-queue-widget" data-company="company-slug"></div><script src="https://inline.app/widget.js"></script></code>	More flexible future embed.
QR code	QR points to company/branch/service join URL.	Download PNG and PDF poster.

Embed Security Requirements

- Embeds must only use public-safe APIs.
- EmbedKey should support allowed domains for JS embed.
- Rate limit join requests from public/embed routes.
- Never expose full customer phone numbers in embed.
- Use CORS and frame policies carefully: allow iframe only for embed routes, not admin routes.
- Embeds must inherit company colors but remain readable and accessible.

10. Payment and Billing Requirements

InLine should monetize through subscriptions, prepaid SMS credits, add-ons, setup/onboarding fees, enterprise invoices, and future runner marketplace commission. Build payment providers as adapters because the best provider may differ by country and company type.

Recommended Launch Payment Strategy

- MVP: manual invoices/EFT + admin mark-as-paid + SMS wallet.
- Next: add online checkout provider adapter for DPO Pay, PayToday, Paystack, or other provider.
- Do not store raw card details. Use hosted checkout/payment links or provider tokenization.
- Enterprise customers should support invoice contracts and manual subscription activation.
- Keep SMS credits separate from subscriptions so heavy SMS users pay more fairly.

Plan	Suggested Price	Includes	Best For
Free	N\$0	1 branch, 1 service, 1 staff, limited tickets, basic public page, trial SMS credits only	Small companies testing the product
Starter	N\$399/month	1 branch, multiple services, 3 staff, QR, SMS enabled, basic analytics, iframe embed	Small businesses
Business	N\$999/month	Multi-branch, more staff, advanced analytics, custom branding, appointments, priority queues	Growing companies
Enterprise	Custom	Unlimited/custom limits, SLA, custom domain, white-label, API access, dedicated onboarding, invoice billing	Banks, hospitals, schools, government offices

Add-ons

Add-on	Suggested Price	Requirement
SMS credits	Prepaid bundles e.g. N\$100, N\$250, N\$500, N\$1000	Each outgoing SMS deducts credits.
Extra branch	N\$150/month per branch	Allows companies to scale.
Extra staff user	N\$50/month per user	Useful for bigger teams.
Custom domain	N\$199/month	company.inline.app or custom domain support.
White-label	Custom pricing	Remove InLine branding for enterprise.
Setup/onboarding	Once-off fee	Implementation help for bigger companies.
API access	Included in Enterprise or paid add-on	For integrations.

Billing Rules

- Subscription status values: TRIALING, ACTIVE, PAST_DUE, CANCELLED, EXPIRED, MANUAL_ACTIVE.
- If subscription is past due, show warning and grace period before locking advanced features.
- Public company page can remain visible when past due, but queue joining may be limited by rules.
- SMS sending must stop when wallet balance is empty unless company is approved for postpaid billing.
- Super admin can override billing with reason, and override must create AuditLog.
- Invoices need invoice number, company, amount, currency, due date, status, payment record, and downloadable PDF later.
- Support annual billing discount later.

11. Dashboards and Analytics

Company Overview Widgets

- Live waiting customers by branch.
- Currently called tickets.
- Average wait time today.
- Customers served today.
- Missed/no-show count.
- SMS balance and low-balance warning.
- Setup checklist completion.
- Plan usage meter: branches, staff, tickets, SMS.

Analytics Feature	Requirement
Wait time analytics	Average, median, longest wait, wait by service, wait by branch.
Service time analytics	Average serving duration, fastest/slowest services.
Peak hour heatmap	Day/time heatmap showing busiest hours.
Staff performance	Served count, average service time, no-show handling.
Customer satisfaction	Ratings and feedback by service/branch.
No-show rate	Missed tickets divided by called tickets.
Queue abandonment	Cancelled tickets and customers who never show.
SMS analytics	Sent, failed, cost, template usage.
Branch comparison	Compare branches side by side.
Forecasting future	Predict tomorrow/next week busy times.

Exports

- CSV export for raw tickets, staff performance, SMS logs, invoices.
- PDF report export for managers.
- Date range filters on every export.
- Exports must respect permissions and tenant isolation.

12. Queue Assistant / Runner Marketplace

This is the powerful future feature: if a company does not use InLine yet, a user can request a verified person to physically stand in line for them. The runner stands in the normal line ethically and updates the user.

Safety rule

This feature must not promote queue cutting, fraud, impersonation, trespassing, fake documents, or breaking company rules. The runner only stands in a normal public line where allowed.

Runner Requirement	Details
Runner verification	ID check, phone verification, profile photo, rating, optional background checks depending on country/legal requirements.
Job request	User enters location, service/reason, desired time, notes, and max budget.
Price estimate	Base fee + waiting time + distance + platform fee.
GPS check-in	Runner checks in at location and periodically confirms presence.
Proof of presence	GPS timestamp, optional photo of queue area where allowed, not private documents.
Position updates	Runner manually updates current position or people ahead.
Masked communication	User and runner can communicate without exposing personal numbers.
Escrow future	Payment held until completed or cancelled based on rules.
Dispute handling	Support can review updates, proof, timestamps, and messages.
Ratings	User rates runner, runner rates job/user behavior.

Runner Buttons

Page	Buttons
Runner Job List	Accept Job, Decline, View Details, Set Availability, Go Offline
Runner Active Job	Check In, Update Position, Upload Proof, Message User, Mark Near Front, Complete Job, Cancel with Reason
User Runner Request	Request Runner, Confirm Rules, Pay, Cancel Request, Message Runner, View Proof, Release Payment, Rate Runner
Admin Runner Moderation	Approve Runner, Reject Runner, Suspend Runner, View Proof, Resolve Dispute, Refund, Release Payment

13. Crazy Cool Features and Future Innovation

These are not all MVP features. They are the "make InLine feel ahead of everyone" ideas. Codex should create placeholder routes/cards/feature flags for them where practical, but build MVP first.

Feature Idea	What It Does
Queue Time Machine	AI predicts when you should leave home based on current queue speed, traffic estimate placeholder, branch distance, and your ticket position.
Ghost Queue Mode	A user can hold a digital place and only physically appear when the system says it is time. Company controls fairness rules.
Queue Health Score	Dashboard score that tells a company whether today is smooth, risky, overloaded, or understaffed.
Crowd Radar	Branch page shows simple crowd level: quiet, normal, busy, packed. Based on queue size and service speed.
Smart Arrival Window	Instead of only ticket number, customer gets a suggested arrival time window: e.g. arrive between 11:20 and 11:35.

Feature Idea	What It Does
Auto Load Balancer	If one branch is busy, suggest another branch nearby with shorter wait.
One-Tap Emergency Priority	For clinics or urgent services: staff can promote urgent cases with mandatory reason and audit log.
SMS Micro-App	Customers with no smartphone can receive short SMS updates and one-way instructions. Two-way support can be provider-dependent.
Voice Queue Assistant	Staff can say "call next" or "mark served" using voice commands in browser future.
TV Mode Themes	Waiting room display with brand colors, announcements, adverts, and next tickets.
Queue Replay	Manager can replay how a day went to see where the line broke down.
Chaos Simulator	Company can simulate busy days: 100 customers, 3 staff, service times, and see predicted bottlenecks.
Customer Mood Signals	Optional quick emoji feedback while waiting: calm, annoyed, confused. Helps staff detect bad experiences early.
Silent Mode for Sensitive Services	Use anonymized ticket codes and hidden service names for privacy-sensitive queues.
Queue Autopilot	AI recommends opening another counter, pausing walk-ins, or sending delay SMS based on real-time bottlenecks.
InLine Passport	Customer profile with saved details and trusted phone verification across many companies.
Universal Queue Card	One clean mobile card that works for any business, with live status and directions.
Company Mini-Site Builder	Public InLine page becomes a small website for businesses without websites.
Smart No-Show Recovery	If a customer misses their turn, system can offer a fair rejoin slot based on company rules.
Ethical Runner Network	Verified runners expand the network before companies install InLine.

14. Data Model

Use PostgreSQL with Prisma. Every table should have id, createdAt, updatedAt where practical. Use UUIDs or cuid. Add indexes on companyId, branchId, queueId, status, date, ticketNumber, phone, slug, and external provider references.

Model	Fields
User	id, name, email, phone, passwordHash, role, status, phoneVerifiedAt, emailVerifiedAt, createdAt, updatedAt
Company	id, ownerId, name, slug, logoUrl, primaryColor, secondaryColor, industry, description, status, createdAt, updatedAt
Branch	id, companyId, name, slug, address, city, phone, openingHoursJson, timezone, isActive, createdAt, updatedAt
Service	id, companyId, branchId optional, name, slug, description, averageServiceMinutes, priorityEnabled, isActive, createdAt, updatedAt
Counter	id, companyId, branchId, name, currentStaffId optional, isActive, createdAt, updatedAt
CompanyStaff	id, companyId, branchId optional, userId, role, permissionsJson, isActive, invitedAt, acceptedAt
Queue	id, companyId, branchId, serviceId, date, status, currentNumber, settingsJson, createdAt, closedAt
QueueTicket	id, companyId, branchId, serviceId, queueId, customerName, customerPhone, ticketNumber, status, source, positionSnapshot, peopleAheadSnapshot, estimatedWaitMinutes, notes, priorityReason, calledAt, servingAt, servedAt, missedAt, cancelledAt, createdAt, updatedAt
TicketEvent	id, ticketId, eventType, actorUserId optional, metadataJson, createdAt
Notification	id, companyId, ticketId optional, userId optional, phone, channel, templateKey, message, status, provider, providerMessageId, cost, errorMessage, createdAt, sentAt, failedAt
Plan	id, name, slug, priceMonthly, priceAnnual, currency, limitsJson, featuresJson, isActive
Subscription	id, companyId, planId, status, billingCycle, currentPeriodStart, currentPeriodEnd, trialEndsAt, cancelAt, provider, providerSubscriptionId
Invoice	id, companyId, subscriptionId optional, invoiceNumber, amount, currency, status, dueDate, paidAt, pdfUrl optional
Payment	id, companyId, invoiceId optional, provider, providerReference, amount, currency, status, paymentMethod, metadataJson, createdAt, paidAt
SmsWallet	id, companyId, balance, currency, lowBalanceThreshold, postpaidEnabled
SmsTransaction	id, companyId, type, amount, balanceAfter, description, notificationId optional, createdAt
EmbedKey	id, companyId, publicKey, allowedDomainsJson, isActive, createdAt, rotatedAt
AuditLog	id, companyId optional, userId optional, action, entityType, entityId, ipAddress, userAgent, metadataJson, createdAt
Runner	id, userId, verificationStatus, rating, isAvailable, currentCity, createdAt, updatedAt
RunnerJob	id, customerId, runnerId optional, locationName, address, reason, status, price, platformFee, runnerPayout, currentPosition, proofUrlsJson, createdAt, acceptedAt, completedAt
RunnerJobUpdate	id, runnerJobId, message, position, latitude optional, longitude optional, photoUrl optional, createdAt

Important Relationships

- Company has many Branches, Services, Staff, Queues, Tickets, Invoices, Payments, Notifications, AuditLogs.
- Branch has many Services, Counters, Queues, Tickets.
- Queue has many QueueTickets.
- QueueTicket has many TicketEvents and Notifications.
- Company has one active Subscription and one SmsWallet.
- Plan limits control number of branches, staff, services, tickets, widgets, analytics, API access.
- RunnerJob is separate from normal QueueTicket because unsupported locations are not controlled by InLine.

15. API Requirements

Auth

- POST /api/auth/register
- POST /api/auth/login
- POST /api/auth/logout
- GET /api/auth/me
- POST /api/auth/forgot-password
- POST /api/auth/reset-password
- POST /api/auth/verify-phone

Companies

- POST /api/companies
- GET /api/companies/:slug
- PATCH /api/companies/:id
- GET /api/companies/:id/dashboard
- GET /api/companies/:id/usage

Branches

- POST /api/companies/:companyId/branches
- GET /api/companies/:companyId/branches
- GET /api/branches/:id
- PATCH /api/branches/:id
- DELETE /api/branches/:id

Services

- POST /api/companies/:companyId/services
- GET /api/companies/:companyId/services
- PATCH /api/services/:id
- DELETE /api/services/:id

Staff

- POST /api/companies/:companyId/staff/invite
- GET /api/companies/:companyId/staff
- PATCH /api/staff/:id
- POST /api/staff/:id/disable
- POST /api/staff/:id/assign-branch

Queues

- POST /api/public/queues/join
- GET /api/public/tickets/:ticketId
- POST /api/public/tickets/:ticketId/cancel
- POST /api/public/tickets/:ticketId/on-my-way
- GET /api/staff/branches/:branchId/queue
- POST /api/staff/tickets/:ticketId/call
- POST /api/staff/tickets/:ticketId/start-serving
- POST /api/staff/tickets/:ticketId/serve

- POST /api/staff/tickets/:ticketId/miss
- POST /api/staff/tickets/:ticketId/hold
- POST /api/staff/tickets/:ticketId/transfer
- POST /api/staff/walk-ins

Embed

- GET /api/embed/:companySlug/config
- POST /api/embed/:companySlug/join
- GET /api/embed/tickets/:ticketId/status

SMS

- GET /api/sms/templates
- POST /api/sms/templates
- PATCH /api/sms/templates/:id
- POST /api/sms/send-test
- GET /api/sms/logs
- GET /api/sms/wallet

Billing

- GET /api/billing/plans
- POST /api/billing/subscribe
- POST /api/billing/create-invoice
- POST /api/billing/mark-invoice-paid
- POST /api/billing/sms/topup
- GET /api/billing/company/:companyId
- POST /api/webhooks/payments/:provider

Analytics

- GET /api/analytics/overview
- GET /api/analytics/wait-times
- GET /api/analytics/staff
- GET /api/analytics/sms
- GET /api/reports/export.csv
- GET /api/reports/export.pdf

Super Admin

- GET /api/admin/companies
- GET /api/admin/revenue
- GET /api/admin/subscriptions
- PATCH /api/admin/companies/:companyId/billing-status
- POST /api/admin/plans
- PATCH /api/admin/plans/:planId
- GET /api/admin/audit-logs

Runner Future

- POST /api/runner/jobs
- GET /api/runner/jobs

- POST /api/runner/jobs/:id/accept
- POST /api/runner/jobs/:id/check-in
- POST /api/runner/jobs/:id/update-position
- POST /api/runner/jobs/:id/complete
- POST /api/runner/jobs/:id/cancel

16. Security, Privacy and Compliance

Area	Requirement
Authentication	JWT access tokens, refresh token strategy, bcrypt password hashing, optional OTP for phone verification.
Authorization	Role-based access control on backend. Tenant isolation on every company-scoped query.
Validation	Zod schemas on all request bodies and query params.
Rate limiting	Public queue join, login, OTP, SMS endpoints, and embed endpoints must be rate limited.
Audit logs	Billing, staff management, queue state changes, payment overrides, and support impersonation must be audited.
PII protection	Mask phone numbers in public views. Never expose customer lists publicly.
Payment safety	Use hosted payment pages/provider tokens. Do not store card details.
SMS consent	Queue join form must disclose that phone number will be used for ticket updates.
Runner safety	No illegal actions, no impersonation, no document handling unless explicitly allowed and legally reviewed.
Backups	Database backups and restore plan required before production.
Logging	Do not log full passwords, OTPs, card data, or sensitive tokens.

Tenant Isolation Checklist

- Every company resource query must include companyId filter.
- Never allow client-provided companyId without verifying membership.
- Super admin access must be explicit and audited.
- Background jobs must include companyId and validate provider configuration.
- Analytics exports must not cross company boundaries.

17. UI/UX Design System

The UI must feel like a premium SaaS product but remain easy enough for a receptionist to use under pressure. Staff console must prioritize speed over decoration. Marketing website can be more visually rich.

Design Area	Requirement
Brand personality	Fast, clean, trustworthy, smart, local-business friendly.
Colors	Dark navy/black for trust, white/light gray background, electric blue or green accent, orange/red warning.
Typography	Use modern readable fonts. Large headings, compact dashboard tables.
Components	Cards, buttons, badges, tables, modals, sidebars, tabs, toasts, empty states, skeleton loaders.
Statuses	WAITING blue/neutral, CALLED orange, SERVING green, SERVED gray, MISSED red, CANCELLED gray, HOLD purple/amber.
Mobile-first	Customer pages and staff console must work well on mobile.
Accessibility	Good contrast, focus states, keyboard navigation, labels on forms.
Animations	Smooth but not slow. Queue updates can pulse lightly.

Reusable Components

- Button, IconButton, LoadingButton, ConfirmDialog, Modal, Drawer, Toast, Badge, StatusBadge.
- DataTable with filters, search, pagination, empty state.
- MetricCard, ChartCard, QueueTicketCard, CurrentTicketCard, BranchCard, ServiceCard.
- PhoneInput, OTPInput, ColorPicker, FileUpload, DateRangePicker.
- PublicCompanyLayout, DashboardLayout, StaffLayout, AdminLayout, EmbedLayout.
- QRCodeCard, EmbedCodeBlock, CopyButton, BillingPlanCard, SmsBalanceCard.

18. MVP Roadmap

Phase	Scope
Phase 0 - Setup	Monorepo, Vite React app, Node API, Prisma, PostgreSQL, Docker Compose, env files, seed data.
Phase 1 - Core Queue MVP	Auth, company onboarding, branches, services, public company pages, join queue, ticket page, staff call next, live Socket.IO.
Phase 2 - SMS and Billing MVP	Mock SMS, SMS wallet, manual invoices/EFT, plans, limits, billing dashboard, SMS logs.
Phase 3 - Embeds and QR	Iframe embed, QR code generator, public service pages, company branding.
Phase 4 - Analytics	Dashboard metrics, wait times, staff performance, exports.
Phase 5 - Provider Integrations	Real SMS provider adapter, payment provider adapter, webhooks, payment links.
Phase 6 - Advanced Features	Appointments, TV mode, kiosk mode, API keys, custom domains.
Phase 7 - Runner Marketplace	Runner verification, jobs, check-ins, proof, payments, disputes.

MVP Must-Have

- A company can sign up and create a branch and services.
- A customer can join a queue from public page without account.
- A customer receives a ticket and can track it live.
- Staff can call next, serve, miss, hold, transfer, and cancel.
- The system sends mock SMS and logs it.
- Company can see billing plan, SMS wallet, and manual invoice state.
- Company can generate QR links and copy iframe embed code.
- Basic analytics show served count, waiting count, average wait, no-show count.

19. Testing and Acceptance Criteria

Test Area	Acceptance Criteria
Queue join	Given branch is open, when customer submits name/phone/service, then ticket is created, number is unique, ticket page opens, SMS job is logged.
Call next	Given waiting tickets exist, when staff clicks Call Next, then first eligible ticket becomes CALLED and live updates are emitted.
Serve ticket	Given ticket is CALLED/SERVING, when staff marks served, then servedAt is set and ticket leaves waiting list.
Billing limits	Given company is on Free plan and reaches limit, when creating extra branch/service/staff, then system blocks with upgrade message.
SMS wallet	Given wallet balance is zero, when SMS event triggers, then SMS is not sent and company sees low balance warning.
Tenant isolation	Given user belongs to Company A, when requesting Company B dashboard API, then response is 403.
Embed	Given iframe embed loads, when customer joins from embed, then ticket source is EMBED and admin can see it.
QR	Given QR link opens, when user joins, then branch/service preselection works.
Audit log	Given staff changes ticket status, then TicketEvent and AuditLog records exist.
Responsive UI	All customer and staff flows must work on mobile width 360px.

Required Empty States

- No branches: show Add Branch button and setup hint.
- No services: show Add Service button.
- No queue tickets: show calm empty queue card.
- No SMS balance: show Buy Credits button.
- No invoices: show no invoices yet.
- No analytics data: show needs more queue activity.
- Closed branch: show closed message and next opening time.

20. Codex Master Build Prompt

Copy the prompt below into Codex when you are ready to generate the project. It references this document and tells Codex how to start implementation.

You are Codex. Build a production-ready full-stack SaaS platform called InLine using React, TypeScript, Vite, Tailwind CSS, Node.js, TypeScript, Express or NestJS, PostgreSQL, Prisma, Socket.IO, JWT auth, bcrypt, Zod validation, and Docker Compose.

InLine is a queueing and virtual ticket platform for companies that do not have ticket systems. It must support public company pages, branch/service queues, QR code joining, embeddable widgets, SMS reminders, company dashboards, staff queue console, billing/subscriptions, SMS wallets, analytics, and a future queue assistant marketplace.

Create a monorepo:

/inline

/apps/web

/apps/api

/packages/shared

docker-compose.yml

README.md

.env.example

Follow this exact build order:

1. Create project structure and shared types.
2. Create PostgreSQL Prisma schema for User, Company, Branch, Service, Counter, CompanyStaff, Queue, QueueTicket, TicketEvent, Notification, Plan, Subscription, Invoice, Payment, SmsWallet, SmsTransaction, EmbedKey, AuditLog, Runner, RunnerJob, RunnerJobUpdate.
3. Add migrations and seed data: super admin, seeded company account, two branches, five services, staff users, plans, sample queue tickets, SMS wallet.
4. Build API with auth, role-based access, tenant isolation, queue joining, ticket tracking, staff queue operations, company setup, billing mock, SMS mock, embeds, analytics basics.
5. Build frontend routes: public marketing pages, login/register, onboarding, public company pages, ticket page, customer dashboard, company dashboard, staff console, embed page, super admin pages.
6. Build core UI components: layouts, buttons, forms, modals, cards, data tables, status badges, queue ticket cards, metric cards, copy button, QR code card.
7. Implement Socket.IO rooms for company, branch, queue, and ticket live updates.
8. Implement MockSmsProvider, SmsProvider interface, SMS logs, SMS wallet deduction, low balance warning.
9. Implement billing models, pricing page, plan limits, manual invoice/EFT workflow, mock payment provider interface.
10. Implement iframe embed first, JS embed placeholder second.
11. Add QR code generation for company, branch, and service join links.
12. Add README instructions to run web, api, database, migrations, seed, and tests.

Important rules:

- Use TypeScript everywhere.
- Use Zod validation on all API input.
- Enforce tenant isolation on backend.
- Hide buttons by permission on frontend, but enforce permissions on backend too.
- Do not store payment card details.
- Public pages must not expose private customer data.
- Every ticket status change must create TicketEvent and AuditLog.

- Every outgoing SMS must create Notification and SmsTransaction.

- Use responsive mobile-first design.

- Build MVP first, but leave clean extension points for payments, SMS providers, runner marketplace, appointments, custom domains, API keys, and white-label.

Design style:

Premium SaaS, dark navy/black, white/light gray, electric blue or green accent, orange/red warnings, clean cards, smooth but fast animations, big staff console buttons, mobile-first customer pages.

Deliver working code, not only skeletons. Prioritize auth, company setup, public join queue, ticket tracking, staff console, SMS mock, billing mock, QR codes, iframe embed, and analytics overview.

21. Sources and Implementation Notes

These sources are implementation notes only. Confirm pricing, contracts, settlement rules, webhooks, compliance, and country support directly with each provider before production launch.

Source	Why It Matters	URL
DPO Pay by Network - Namibia online payments	DPO states its Namibia solution supports Namibian merchants and settlement in NAD. Useful as a possible online payment adapter.	https://dpogroup.com/online-payments/namibia/
PayToday Namibia - business gateway	PayToday advertises a payment gateway for eCommerce and a WooCommerce payment plugin. Useful for local payment exploration.	https://site.paytoday.com.na/
Twilio Namibia SMS Guidelines	Twilio lists Namibia SMS guidelines and shows two-way SMS supported as No, so design SMS as primarily one-way unless another provider supports replies.	https://www.twilio.com/en-us/guidelines/na/sms
Africa's Talking Pricing	Africa's Talking pricing page includes Namibia under SMS coverage/pricing information. Useful as an SMS provider to test.	https://africastalking.com/pricing

Final Build Notes

- Start simple, but keep architecture expandable.
- Manual invoice/EFT can get paying companies onboard before full payment integration is finished.
- SMS must be treated as paid usage from the beginning, even if the first version uses mock SMS.
- Embed-first is a major sales advantage for companies with existing websites.
- The runner marketplace should be feature-flagged and launched only after legal, safety, and operational review.